$\gamma: \underbrace{9x^2+24xy+16y^2}-20x+15y=0$ γ NON deg

Caso $a_{12}\neq 0$ Dobbiamo trovare rotaz. P t.c. ${}^tP\cdot A P=\begin{bmatrix}\lambda_1 & 0\\ 0 & \lambda_2\end{bmatrix}$, dove

$A=\begin{bmatrix}9 & 12\\ 12 & 16\end{bmatrix}$ $\quad p(\lambda)=|A-\lambda I|=\begin{vmatrix}(9-\lambda) & 12\\ 12 & (16-\lambda)\end{vmatrix}=$ (γ parabola)

$=\lambda^2-25\lambda+144-144=\lambda(\lambda-25)$ $\quad \lambda_1=25$ $\lambda_2=0$

Cerchiamo P t.c. ${}^tP\cdot AP=\begin{bmatrix}25 & 0\\ 0 & 0\end{bmatrix}$ $\quad P=[\vec{v}_1,\vec{v}_2]=$

$\vec{v}_1$ si ricava da:

$\underbrace{(A-\lambda_1 I)}_{\lambda=1}\begin{bmatrix}x\\ y\end{bmatrix}=\begin{bmatrix}0\\ 0\end{bmatrix}$, cioè $\begin{cases}-16x+12y=0\\ 12x-9y=0\end{cases}$ $\quad \vec{v}_1=\text{vers}\left(\begin{bmatrix}3\\ 4\end{bmatrix}\right)=$

$=\begin{bmatrix}3/5\\ 4/5\end{bmatrix}$

$4x=3y$

$\vec{v}_2$ --- $\quad \vec{v}_2=\begin{bmatrix}-4/5\\ 3/5\end{bmatrix}$ $\quad P=\begin{bmatrix}3/5 & -4/5\\ 4/5 & 3/5\end{bmatrix}$

$\begin{bmatrix}x\\ y\end{bmatrix}=P\begin{bmatrix}x'\\ y'\end{bmatrix}\iff \begin{cases}x=\frac{1}{5}(3x'-4y')\\ y=\frac{1}{5}(4x'+3y')\end{cases}$

Possiamo scrivere eq. di γ rispetto alle nuove:

$\gamma: 9x^2+24xy+16y^2-20x+15y=0$

$\lambda_1 x'^2+\lambda_2 y'^2$

$=25x'^2+0y'^2$

$\gamma: 25x'^2-12x'+16y'+12x'+9y'=0$

$\gamma: y'=-x'^2$

$A=\begin{bmatrix}a_{11} & a_{12}\\ a_{12} & a_{22}\end{bmatrix}$ $\quad \det A=\lambda_1\lambda_2$

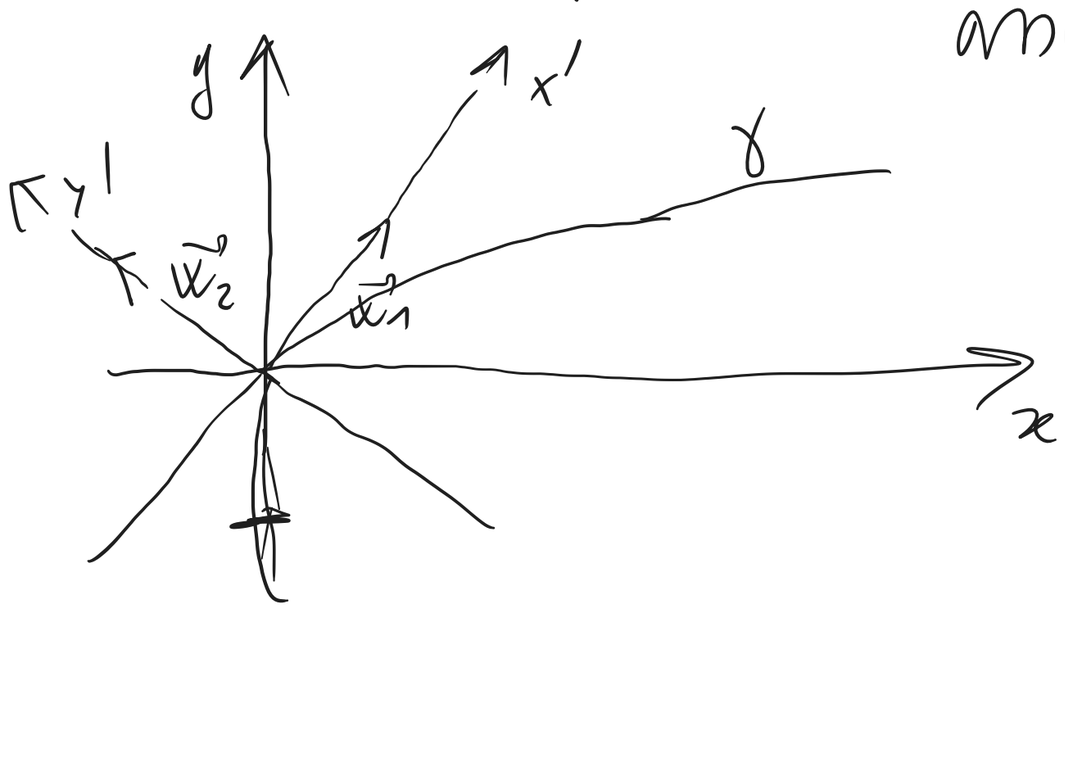
$\det A>0$ ellisse

$\det A=0$ parabola (scelta ovviamente) $\lambda_2=0$ $\lambda_1\neq 0$

$\det A<0$ iperbole

$P=\begin{bmatrix}3/5 & -4/5\\ 4/5 & 3/5\end{bmatrix}=[\vec{v}_1,\vec{v}_2]$

asse x': $4x-3y=0$



Determinare intersez. γ con assi di partenza.